

core NumPy array creation methods Simple Hinglish. Real-life analogies.

1. `zeros()`

Kya karta hai: Saare elements ko **0** se bhara hua array banata hai.

Real life analogy: Socho ek **attendance sheet** jisme abhi kisi ka attendance mark nahi hua. Sab zero.

Example:

```
import numpy as np
np.zeros(5)
```

Output: `[0. 0. 0. 0. 0.]`

Use case:

- Counters
- Scores initialize karna
- Empty sensor readings

2. `ones()`

Kya karta hai: Saare elements ko **1** se bhara hua array banata hai.

Real life analogy: Ek **factory switchboard** jahan sab machines ON (1) state me hain.

Example:

```
np.ones(4)
```

Output: `[1. 1. 1. 1.]`

Use case:

- Default weights
 - Flags
 - Masking operations
-

3. `full()`

Kya karta hai: Array jisme **har element same fixed value** hota hai (0 ya 1 zaroori nahi).

Real life analogy: Class ke sab students ko **same bonus marks = 5** mile.

Example:

```
np.full(6, 37)
```

Output: `[37 37 37 37 37 37]`

Use case:

- Constant temperature assumption
 - Default fill values
 - Padding data
-

4. `arange()`

Kya karta hai: Ek **sequence** banata hai, jaise Python `range()` but NumPy speed ke saath.

Real life analogy: Train station ke **platform numbers:** 0, 1, 2, 3, 4...

Example:

```
np.arange(0, 10, 2)
```

Output: `[0 2 4 6 8]`

Meaning: `start, stop, step`

Use case:

- Time steps
 - Indexes
 - Simulation loops
-

5. `eye()`

Kya karta hai: **Identity matrix** banata hai → diagonal = 1, baaki sab = 0

Real life analogy: School ke **result sheet** me sirf apna roll number apni row se match karta hai. Baaki sab mismatch (0).

Example:

```
np.eye(4)
```

Output:

```
[[1.  0.  0.  0.]  
 [0.  1.  0.  0.]  
 [0.  0.  1.  0.]  
 [0.  0.  0.  1.]]
```

Use case:

- Linear algebra
- ML models
- Transformations
- Neutral element (like multiplying by 1)

One-line mental model (important)

- **zeros** → **empty start**
- **ones** → **all active**
- **full** → **constant assumption**
- **arange** → **ordered sequence**
- **eye** → **self-mapping / identity**

If you understand these 5 properly, you already know **80% of NumPy array creation** used in real systems.